

SUMAN MANDAL

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PROFESSIONAL SUMMARY

Results-driven **Java Backend Developer** and final-year B.Tech Computer Science student with hands-on experience designing and deploying **production-ready RESTful APIs** using **Java 21** and **Spring Boot 3.5**. Proficient in **MongoDB, Docker, Kubernetes**, and **CI/CD pipelines** via GitHub Actions. Experienced in **JWT-based authentication, Spring Security**, role-based access control, and microservices architecture. Passionate about building scalable, high-performance backend systems with clean code and comprehensive test coverage using **JUnit 5** and **Mockito**.

TECHNICAL SKILLS

Languages:	Java 21, Python, SQL
Frameworks & Libraries:	Spring Boot 3.5, Spring Security, Spring Data MongoDB, Spring MVC, Hibernate Validator, JWT, Thymeleaf, Lombok
Backend & API:	RESTful API Design, JWT Authentication, Role-Based Access Control (RBAC), OAuth2, Microservices Architecture, API Gateway
Databases:	MongoDB Atlas, MySQL, PostgreSQL, Redis
DevOps & Cloud:	Docker, Docker Compose, Kubernetes (K8s), Helm, GitHub Actions CI/CD, Prometheus, Grafana
Testing:	JUnit 5, Mockito, k6 Load Testing, Postman, Integration Testing
Tools & Practices:	Git, GitHub, Maven, IntelliJ IDEA, Swagger/OpenAPI, Structured Logging (Logback), SonarQube
Core Concepts:	Data Structures & Algorithms, OOP, System Design, Design Patterns, SOLID Principles

PROJECT EXPERIENCE

Society Management System — Production-Ready Spring Boot Backend

Technologies: Java 21, Spring Boot 3.5, Spring Security, MongoDB Atlas, Docker, Kubernetes, GitHub Actions, Prometheus, Grafana, JUnit 5, Mockito, k6

GitHub: github.com/RajsProjects/society-management

- Architected and developed a **production-ready RESTful API** with **5 core modules**: User Management, Issue Tracking, Maintenance Billing, Announcements, and Email Notifications.
- Implemented **JWT-based stateless authentication** with Spring Security, enforcing **role-based access control (RBAC)** for ADMIN and RESIDENT roles across all endpoints.
- Designed a **resident approval workflow** (PENDING → ACTIVE → INACTIVE) with admin-controlled status management and email notification triggers via Spring ApplicationEvent.
- Built **maintenance billing module** with UPI payment simulation, automatic overdue detection using **@Scheduled cron jobs**, and async email notifications via JavaMailSender + Thymeleaf templates.
- Implemented **community issue tracking** with vote integrity enforcement using compound MongoDB indexes, preventing duplicate votes at the database level.
- Achieved **78 unit tests** with JUnit 5 and Mockito covering all service layers; optimized N+1 query issues in issue listing with batch repository calls.
- Containerized the application with a **multi-stage Dockerfile** and deployed using **Docker Compose** with Prometheus + Grafana observability stack.
- Configured **Kubernetes manifests** including Deployment, Service, HorizontalPodAutoscaler (HPA), ConfigMap, and Secrets for production-grade K8s deployment.

- Set up **GitHub Actions CI/CD pipeline** with automated test execution, Docker image build, and push to Docker Hub on every merge to main branch.
- Conducted **k6 load testing** with 200 concurrent virtual users; achieved **0% error rate** and **100% check success rate** across 17,922 HTTP requests.
- Documented all APIs using **Springdoc OpenAPI / Swagger UI** with JWT Bearer authentication, @Operation annotations, and structured error responses.
- Implemented **global exception handling** with structured JSON error responses, MDC-based correlation IDs for distributed tracing, and environment-based logging profiles.

EDUCATION

Bachelor of Technology — Computer Science (Artificial Intelligence)

Sri Balaji College of Engineering and Technology, Jaipur | Graduation: 2026

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, System Design, Object-Oriented Programming, Generative AI, Data Mining

CERTIFICATIONS & ACHIEVEMENTS

- Software Engineer Intern Certification — HackerRank (2025)
- Java Programming Certification | System Design Certification | Python Programming Certification
- Campus Mantri (Campus Ambassador) — GeeksforGeeks (2026)
- 1st Position — Project Display Competition | Runner-Up — College Hackathon